Windows PowerShell

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Windows PowerShell 5.1
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PS C:\Users\TEA\Downloads\nist-crypto-demo> python --version
Python 3.13.1
PS C:\Users\TEA\Downloads\nist-crypto-demo> python -m pip install -r requirements.txt
Collecting cryptography (from -r requirements.txt (line 1))
  Downloading cryptography-49.0.0-cp311-abi3-win_amd64.whl.metadata (4.3 kB)
Collecting pycryptodome (from -r requirements.txt (line 2))
  Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl.metadata (3.5 kB)
Collecting cffi>=2.0.0 (from cryptography->-r requirements.txt (line 1))
  Downloading cffi-2.1.0-cp313-cp313-win_amd64.whl.metadata (2.6 kB)
Collecting pycparser (from cffi>=2.0.0->cryptography->-r requirements.txt (line 1))
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Downloading cryptography-49.0.0-cp311-abi3-win_amd64.whl (3.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 4.3 MB/s eta 0:00:00
Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 3.0 MB/s eta 0:00:00
Downloading cffi-2.1.0-cp313-cp313-win_amd64.whl (185 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Installing collected packages: pycryptodome, pycparser, cffi, cryptography
Successfully installed cffi-2.1.0 cryptography-49.0.0 pycparser-3.0 pycryptodome-3.23.0

[notice] A new release of pip is available: 24.3.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
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PS C:\Users\TEA\Downloads\nist-crypto-demo> python app.py
======================================================================
 IMPLEMENTATION OF NIST SECURITY STANDARDS
 Student: Tahniat Farhan | Reg No: 2025(S)-CYS-15 | UET Lahore
======================================================================


======================================================================
1. DIFFIE-HELLMAN KEY EXCHANGE (NIST SP 800-56A)
======================================================================
  -> Generated 2048-bit DH domain parameters
  -> Alice and Bob each generated a private/public key pair
  Alice derived key (hex): 4cf8e8e0c469b637c74c99efdded7815...
  Bob   derived key (hex): 4cf8e8e0c469b637c74c99efdded7815...
  Keys match: True
  RESULT: PASS - Shared secret established securely

======================================================================
2. RSA-2048 KEY PAIR GENERATION (NIST FIPS 186-4)
======================================================================
  -> RSA-2048 key pair generated
  -> Saved keys/rsa_private.pem and keys/rsa_public.pem
  RESULT: PASS

======================================================================
3. DIGITAL SIGNATURE - RSA-PSS / SHA-256 (FIPS 186-4)
======================================================================
  Message   : Implementation of NIST Security Standards - Tahniat Farhan
  Signature : buXI6IejpLrrByO1xGFC0TTy9ROVr0o73cfa9nRsDF8CRHuTAWoSntzfZKR4...
  Signature verification (unmodified message): True
  Tampered message correctly rejected: True
  RESULT: PASS
```

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======================================================================
4. DIGITAL CERTIFICATE - X.509 SELF-SIGNED (PKI)
======================================================================
  -> Self-signed X.509 certificate created and saved to certs/certificate.pem
  Subject      : CN=Tahniat Farhan,O=UET Lahore - Cyber Security Dept,L=Lahore,ST=Punjab,C=PK
  Serial Number: 258865755065453271609507127557611360861652484402
  Valid From   : 2026-07-11 14:55:44+00:00
  Valid Until  : 2027-07-11 14:55:44+00:00
  Certificate signature valid: True
  RESULT: PASS


======================================================================
5. AES-256-GCM SYMMETRIC ENCRYPTION (NIST FIPS 197)
======================================================================
  Plaintext  : Confidential Cryptography Lab Report - UET Lahore
  Ciphertext : 026e8312a2bed3c927be014a04b92b44c318db2f3474e39440...
  Auth Tag   : a76992f0aa780bf6a58b47d4d8721388
  Decrypted  : Confidential Cryptography Lab Report - UET Lahore
  RESULT: PASS


======================================================================
6. TRIPLE DES (3DES) ENCRYPTION (NIST SP 800-67, legacy)
======================================================================
  Plaintext  : Legacy symmetric cipher demo
  Ciphertext : d641dc0e28d9ddb57b6ffdb5aea9d56ade2948f6958986362fa801555dc9b66d
  Decrypted  : Legacy symmetric cipher demo
  Note: 3DES is deprecated by NIST (disallowed after 2023) and
        is included here only for syllabus/legacy comparison.
  RESULT: PASS
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======================================================================
7. HYBRID ENCRYPTION - RSA-OAEP WRAPS AES KEY (Basis of TLS)
======================================================================
  RSA-encrypted AES key: 627dbd83c7adf8880bd4579123e37df1d72a0d40cf0d3e0764...
  Recovered AES key matches original: True
  RESULT: PASS

======================================================================
8. ONE-TIME PAD (OTP) ENCRYPTION
======================================================================
  Original  : OTP_IS_UNBREAKABLE
  Key (hex) : fea76ef2528c94d1fb2d48dd065511fe760a
  Cipher    : b1f33ead1bdfcb84b56f1a98471e50bc3a4f
  Decrypted : OTP_IS_UNBREAKABLE
  RESULT: PASS

======================================================================
SUMMARY OF RESULTS
======================================================================
  [PASS] Diffie-Hellman
  [PASS] Digital Signature
  [PASS] Digital Certificate
  [PASS] AES-256-GCM
  [PASS] 3DES
  [PASS] Hybrid RSA+AES
  [PASS] One-Time Pad
======================================================================
 ALL NIST SECURITY STANDARDS IMPLEMENTED SUCCESSFULLY!
======================================================================
PS C:\Users\TEA\Downloads\nist-crypto-demo>
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